



MAHMUT ESAT KOLAY

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EDUCATION



Selcuk University

Sep 2023 - Jun 2027

Bachelor of Computer Engineering

- Working on projects in Machine Learning, Deep Learning, and Data Science using Python and real-world datasets
- Studied fundamental topics: algorithms, data structures, object-oriented programming, databases



Mehmet Tuza Pakpen Technical High School

Sep 2019 - Jun 2023

Industrial Automation Technologies

- Worked on IoT-based projects combining basic software and electronics skills
- Studied core topics: microcontroller programming, circuit design, PLC systems, and industrial sensors

PROJECTS

AI-Driven Inventory Optimization & Demand Forecasting

- Developed an end-to-end predictive analytics framework using XGBoost to optimize FMCG raw material inventory; engineered time-series features and a dynamic safety stock algorithm that reduced total inventory costs by 32.6% while maintaining a 95% service level against simulated demand volatility.

Smart OEE Dashboard & Production Loss Analytics

Tech: Python, Pandas, XGBoost, Streamlit, Plotly

- Analytics Engine:** Developed an end-to-end OEE tracking engine for high-volume FMCG lines, converting raw data into actionable KPIs (Availability, Performance, Quality).
- Loss Diagnostics:** Built a "Six Big Losses" module with automated Pareto analysis to identify critical downtime causes, driving operational agility.
- Interactive Dashboard:** Created a real-time Streamlit UI for machine/shift-based drill-down analysis in high-speed industrial environments.
- Predictive Risk Modeling:** Implemented an XGBoost model to forecast shift-based OEE risks, enabling proactive intervention via historical trend analysis.

Agentic RAG Research Assistant Technologies:

Developed an end-to-end RAG system for academic PDFs using hierarchical semantic chunking. Achieved 80%+ relevance via Hybrid Search (BM25+Vector) and Cross-Encoder re-ranking. Integrated a custom MCP server for autonomous local file management and research note-taking.

Aerospace Battery RUL Prediction Technologies: Deep Learning, LSTM, Explainable AI

- Developed an explainable LSTM + attention-based RUL model for aerospace Li-Ion batteries, achieving ~84% RMSE reduction in critical cases and validated across multiple thermal conditions; selected for T3 Foundation Young Researcher Support.

6-DOF Robot Arm Forward Kinematics Simulator Technologies: Python, NumPy, Robotics Library

- Developed a simulator to calculate and visualize the end-effector position of a 6-DOF robotic arm using forward kinematics equations.

COMPETITIONS

Unilever Chain Reaction @ U-House

- Participated in a prestigious supply chain simulation; conducted data-driven optimization strategy analyses in a competitive environment.

TÜBİTAK 2209-A Undergraduate Research

- Developed an image processing-based optimization project for the detection and classification of oilseeds using deep learning methods.

Emlak Konut - Smart Vertical Transport & Robotic Construction Comp.

- Ranked among the top 10 teams and qualified for the finals.

BTK Ankara AI Hackathon @ Ankara Teknopark

- As part of a team, designed a functional AI-based prototype application within 30 hours under real-world constraints.

WORK EXPERIENCE



Intern - Computer Engineering & Product Management

Sep 2025 - Nov 2025

- Completed an AI & Startup Strategy Fellowship, applying Founder and Project Mindset to develop ideas from 0 to 1.
- Built workflow automations with UiPath and used Google AI tools to create generative AI-driven business solutions.



Intern - Computer Engineering & Robotics

June 2025 - July 2025

- Designed and implemented ML-driven fault detection algorithms for industrial robotic systems
- Programmed KUKA robots and analyzed motion through kinematics and dynamic modeling



Intern - Industrial Automation & Electrical

June 2023 - July 2023

- Participated in the installation, maintenance, and troubleshooting processes of factory electrical systems.
- Conducted assembly, fault detection, and monitoring operations for power factor correction (PFC) panels.
- Actively contributed to the company's specialized compensation panel projects, ensuring energy efficiency and system stability.



Intern - Industrial Automation & Control Systems

June 2022 - Oct 2022

- Assisted in PLC programming, control panel design, and machine wiring for industrial automation systems
- Contributed to the development of electrical systems for packaging and stretch wrapping machines
- Observed and analyzed existing automation systems to identify improvements in operational efficiency

CERTIFICATES



TEI Aviation Engines School

- TUSAS Engine Industries | Jan 2026 - Present
- (Accepted into the specialized aerospace engine training program)



Unilever Supply Chain + Chain Reaction



Aspire Institute - Aspire Leaders Program



Deloitte Australia - Data Analytics



Fundamentals of Deep Learning



Academy Of Responsible Development Erasmus+



Intermediate Machine Learning

TECHNICAL SKILLS

Data Eng. & Analytics: Power BI, SQL, ETL, Web Scraping, Pandas

AI & Machine Learning: LLM, RAG, Computer Vision, Deep Learning

Languages: Python, C++, C#, Java

Robotics & Automation: KUKA Sim Pro, KRL, PLC, ROS, IoT

DevOps & Tools: Git, Docker, UiPath, Linux